

RFR Regime

(8. Multi Currency Curves)

FX Quote



- ▶ Understanding FX Quote is crucial to get the conversion rate right. The first currency is called the **base** currency, and the second currency is called the **quote** currency.

$$\begin{array}{ccc} \text{EUR} & / & \text{USD} = 1.1250 \\ \underbrace{\quad} & & \underbrace{\quad} \\ \text{(base currency)} & & \text{(quote currency)} \end{array} \quad (1 \text{ EUR} = 1.1250 \text{ USD})$$

- ▶ There are two popular Quote Conventions. The FX spot and futures can be quoted in different conventions for the same currency pair.

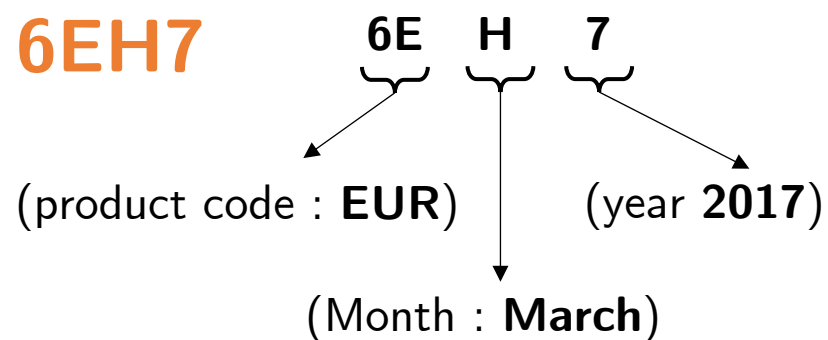
▶ European Quote **USD / EUR , USD / JPY , USD / CHF** (USD = base currency)

▶ American Quote **EUR / USD , JPY / USD , CHF / USD** (USD = quote currency)

CME FX Futures Product Code and Quotes



- ▶ All currency futures on CME Globex are listed using the following trading code convention.



JAN	FEB	MAR	APR	MAY	JUN
F	G	H	J	K	M
JUL	AUG	SEP	OCT	NOV	DEC
N	Q	U	V	X	Z

- ▶ For example, the EUR/USD is quoted as USD per EURO on CME's website

<https://www.cmegroup.com/markets/fx/g10/euro-fx.quotes.html>

<https://www.cmegroup.com/rulebook/CME/>

CME FX Futures Contract Specification

	EUR/USD	JPY/USD	GBP/USD	CAD/USD	AUD/USD	MXN/USD
PRODUCT	Euro	Japanese Yen	British Pound	Canadian Dollar	Australian Dollar	Mexican Peso
PRODUCT SYMBOL	6E	6J	6B	6C	6A	6M
CONTRACT SIZE	125,000 euro	12.5 million yen	62,500 pounds	100,000 Canadian dollars	100,000 Australian dollars	500,000 pesos
EXPIRATIONS	Quarterly, monthly	Quarterly, monthly	Quarterly, monthly	Quarterly, monthly	Quarterly, monthly	Quarterly
MINIMUM TICK SIZE	\$0.00005 per euro increments	\$0.0000005 per yen increments	\$0.0001 per pound increments	\$0.00005 per CAD increments	\$0.00005 per AUD increments	\$0.00001 per peso increments
DOLLAR VALUE OF A TICK	\$6.25	\$6.25	\$6.25	\$5.00	\$5.00	\$5.00

Dollar value of a tick = Contract size x Minimum Tick size

Notional Value = Contract size x Quote

PnL = Contract size x (Δ Quote)

FX MARKET SNAPSHOT >

6EM5 OPT  Euro FX Futures 1.1397 169,781 -0.00175 (-0.15%)	6JM5 OPT  Japanese Yen Futures 0.006998 173,554 -0.0000485 (-0.69%)
6BM5 OPT  British Pound Futures 1.3313 63,505 -0.0016 (-0.12%)	6CM5 OPT  Canadian Dollar Futures 0.7237 48,315 +0.0003 (+0.04%)
6MM5 OPT  Mexican Peso Futures 0.050930 29,524 +0.000230 (+0.45%)	6NM5 OPT  New Zealand Dollar Futures 0.5971 24,732 -0.00295 (-0.49%)

CME FX Futures Settlement



- ▶ Standard Futures settle on the **third Wednesday** (IMM dates) and trading terminates the **second business day preceding the third Wednesday** of the contract month (usually Monday)
- ▶ The highest volumes and tightest liquidity typically occur in the nearest quarterly contract month
- ▶ As the last trading day approaches, a trader typically has three choices
 - ▶ Close out the position by entering an opposite trade leaving with no Euro exposure
 - ▶ Roll the future to next expiring quarterly contract month to maintain the EUR exposure. For example, if a trader is long March contract, they can sell **Mar/Jun spread** (sell Mar, buy Jun) to create a long Jun position.
 - ▶ Make Delivery (Short) or Take delivery (long) of the currencies. For example, in case of EUR futures, short positions will deliver 125,000 euros to the long position per standard contract

GBP Futures Settlement Example



6BH5

- ▶ March contract **6BH5** has Expiry Date = 19th March, Last Trading Date = 17th March
- ▶ Monday (19th March) **6BH5** quote = 1.2405 and deferred contract June **6BM5** quote = 1.2395
- ▶ **Calendar spread = Contract Quote – Deferred Quote = 1.2405 - 1.2395 = 0.0010**
- ▶ In the final 30 seconds of trading CME calculates VWAP of the deferred contract (assume 1.2390)
- ▶ Final Settlement price = 1.2390 + 0.0010 = 1.2400

FX Link



- ▶ CME FX link provides a seamless connection between the OTC spot market and the CME future market. The FX link is a spread between FX Future and FX Spot. This is equivalent to the **Forward points or swap points** which can be +ve or -ve.

$$\text{FX Link Spread} = \text{CME Future price} - \text{OTC Spot price}$$

- ▶ If the future and spot quotes follow the same convention, the FX link is called **non-inverted**, else it's called **inverted**. It's an important consideration to get the calculations right.
- ▶ For **non-inverted** currencies, **long** on the FX link spread means **buying futures** and **selling spot**. For **inverted** currencies, **long** on FX link spread means **buying spot** and **selling futures**.

Covered Interest Parity (CIP)

- ▶ CIP is like a physical law in international finance which works on no arbitrage principle. The **implied future** price is linked to the spot price through the short-term interest rate differential of the two currencies and the days until expiry. This result is called **Covered Interest rate parity** (CIP).

$$\text{Future Price}(F) = \text{Spot price}(S) \times \left(\frac{1 + R_{\text{quote}} \left(\frac{d}{360} \right)}{1 + R_{\text{base}} \left(\frac{d}{360} \right)} \right)$$

- ▶ If $R_{\text{quote}} < R_{\text{base}}$ (positive carry), Then $F < S$ (Future trades at discount) and basis < 0
- ▶ if $R_{\text{quote}} > R_{\text{base}}$ (negative carry) , Then $F > S$ (Future trades at premium) and basis > 0

xCy basis

- ▶ Due to market frictions and specific demand and supply conditions, CIP does not hold in practice. The traded future prices will be slightly different from what is expected from CIP. The difference between the traded price and the CIP price is called as **cross-currency basis**.
- ▶ The CME EUR/USD Cross Currency Basis Index (**XEURB**) measures the basis point deviation from the covered interest parity implied forward FX price. This is the difference between the next ESTR quarterly IMM futures interest rate and the implied ESTR interest rate from the next Euro FX Future and SOFR quarterly IMM futures. It can be calculated using the formula below

$$\text{xCy basis} = \underbrace{\left(\frac{360}{d} \right) \left[\left(\frac{S}{F} \right) \left(1 + R_{SOFR} \left(\frac{d}{360} \right) \right) - 1 \right]}_{\text{(implied ESTR from FX Future)}} - \underbrace{R_{ESTR}}_{\text{(observed ESTR)}}$$

Handwritten annotations: A red $\frac{1}{\delta}$ is above the $\frac{360}{d}$ term. A red δ is above the $\frac{d}{360}$ term. A red $r^{\$}$ is above the R_{SOFR} term. A red δ is above the R_{ESTR} term.

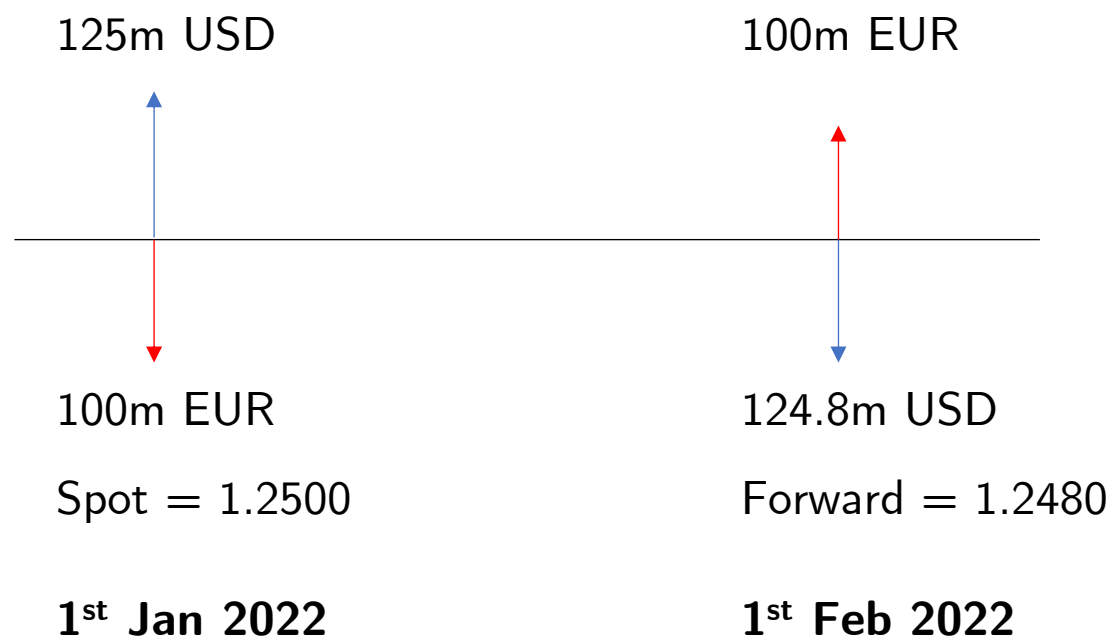
<https://www.cmegroup.com/markets/interest-rates/stirs/eur-xccy.settlements.html>

<https://www.cmegroup.com/markets/interest-rates/cme-group-cross-currency-basis-watch.html>

FX Swap

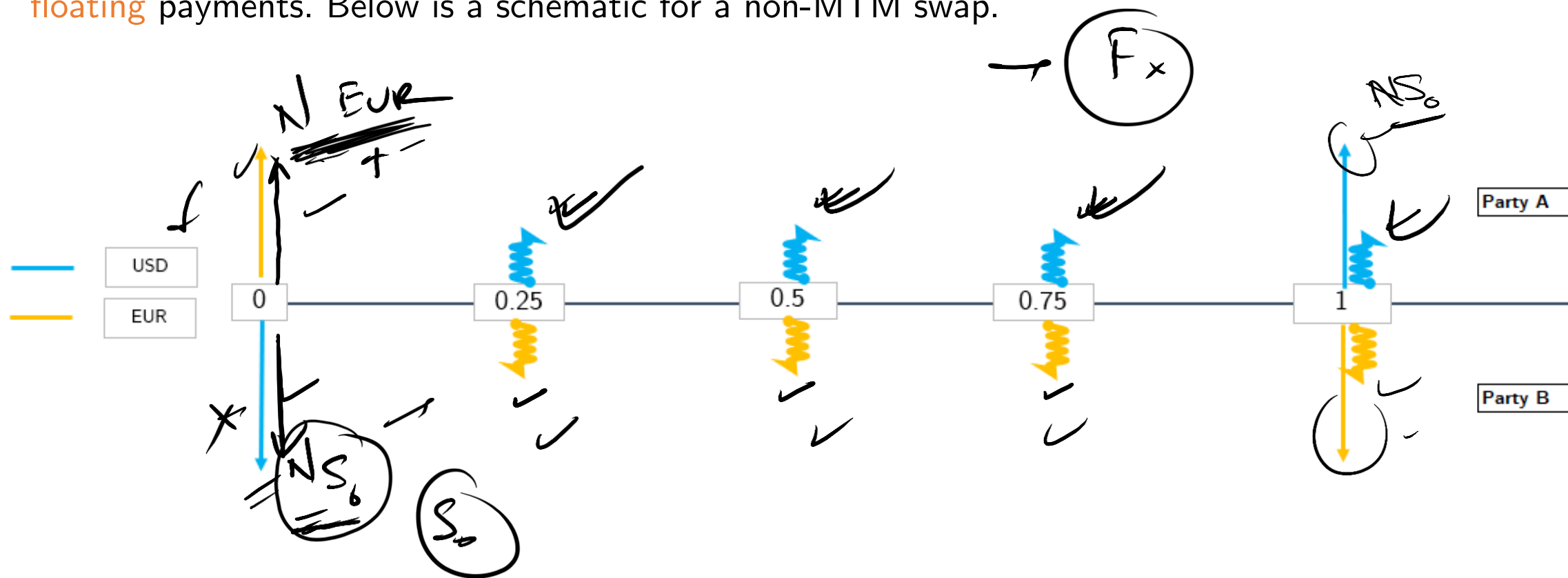


- ▶ FX Swaps are agreements to complete two offsetting exchanges: the "near leg" is the initial exchange of two currencies at the spot rate, and the "far leg" is the reverse exchange of the same currencies at a predetermined forward rate.



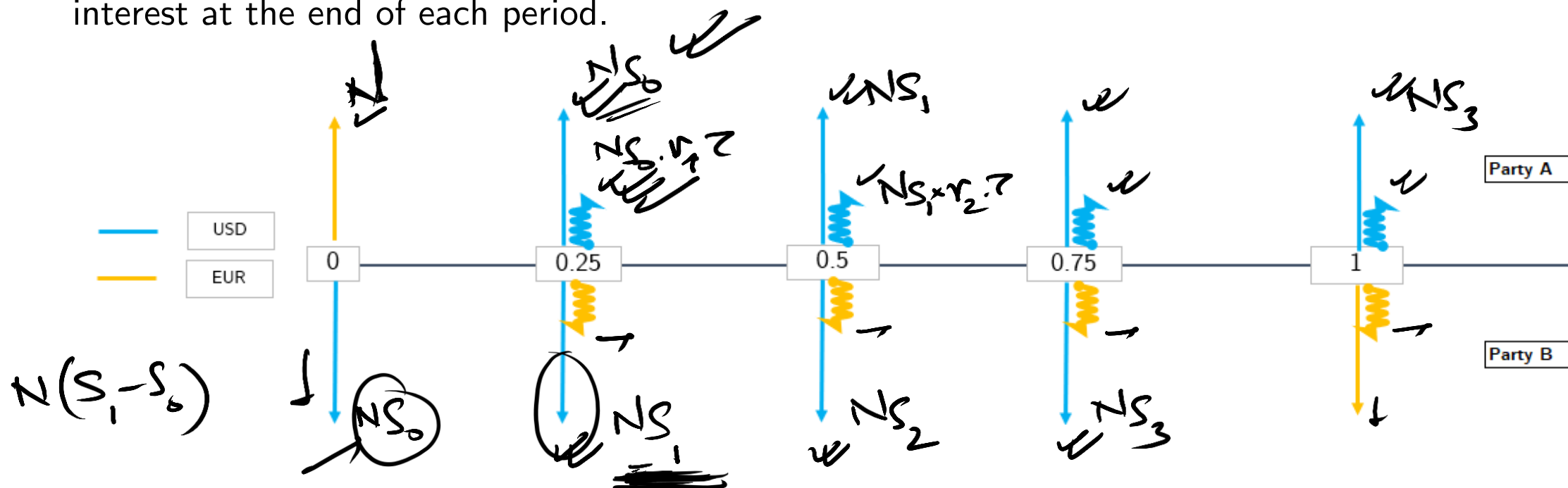
Non-MTM XCCY

- A non-MTM cross-currency swap involves exchanging nationals in two different currencies at the beginning and at the end using the same rate as the beginning and periodic interest payments (fixed or floating). The contract can be customized to have **fixed-fixed**, **fixed-floating** or **floating-floating** payments. Below is a schematic for a non-MTM swap.



MTM XCCY

- ▶ An MTM cross-currency swap is the most popular variant of cross-currency swaps. Typically, the non-USD notional is kept fixed, whereas the USD notional gets adjusted based on prevailing spot rates on coupon dates. This is to limit the exposure and credit risk.
- ▶ The adjustment leg (USD) represents a series of single period loans and payment of principal with interest at the end of each period.



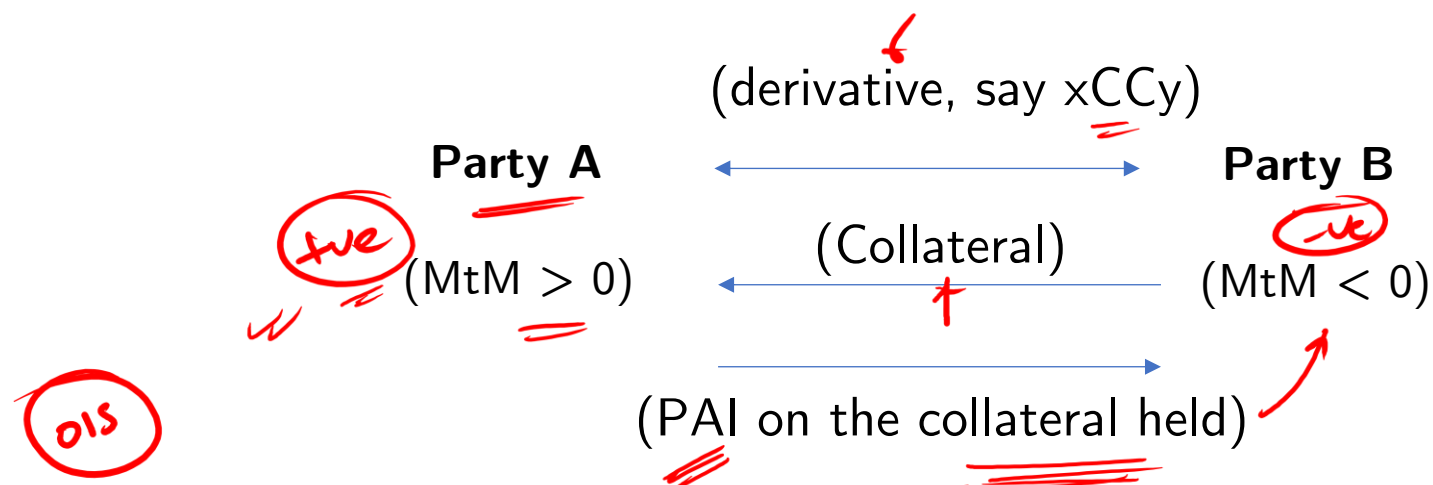
Bloomberg MTM XCCY Swap

- Notice the spread is -24.500 basis points added to EUR leg which makes the swap NPV = 0



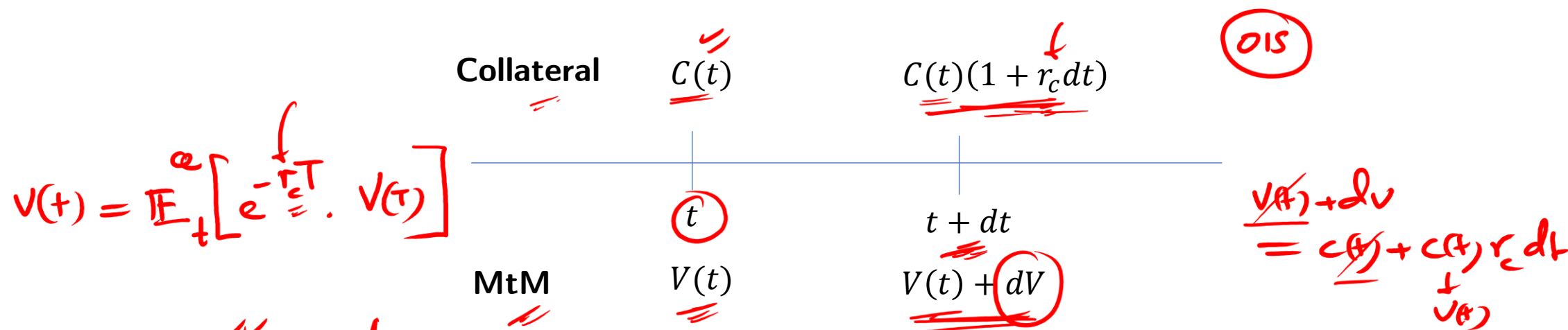
CSA Collateral Basics

- ▶ Credit Support Annex (CSA) is a part of the ISDA Master Agreement which specifies the details of collateral arrangements, remuneration, haircuts, currency options etc. Collaterals reduce the net exposure and hence counterparty credit risk.
- ▶ In the example below, say A and B get into a cross-currency swap. The valuation of the swap from A's perspective is +ve. Then B deposits collateral to A to offset the exposure. A pays remuneration to B for holding the collateral which is called as Price Alignment Interest (PAI)
- ▶ LCH and CME recently switched from OIS (EFFR) rate to SOFR rate for PAI on USD collaterals



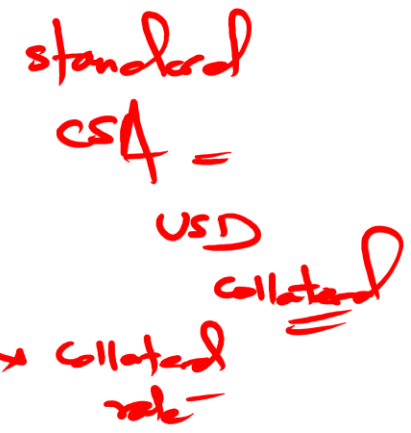
CSA Collateral – Single Currency

- ▶ Can we use the same discounting curve for uncollateralized and collateralized cashflows ? The answer is no. It follows from no-arbitrage principles that derivative needs to be priced using the same rate that is earned by the collateral.
- ▶ Consider a EUR derivative position collateralized by EUR cash. We will assume no frictions, and that collateral always matches the MtM of the position.



It follows that $dV = V(t)r_c dt$, so the appropriate discount factor to discount EUR cashflow collateralized by EUR is the EUR O/N collateral rate or ESTR (earlier EUR OIS)

- \$
C
C
C



CSA Collateral – Effective Funding Rate

- Note that the EUR funding rate in presence of USD collateral is not the EUR RFR rate. This is because of the basis present in the xCCy market. The implied EUR funding rate is given as

$$\frac{1}{\delta} \left[\left(\frac{C^{\$}(t)(1 + r^{\$}\delta)}{F(0, \delta)} \right) / (C^{\$}(t)/S_0) - 1 \right] = \frac{1}{\delta} \left[\left(\frac{S_0(1 + r^{\$}\delta)}{F(0, \delta)} \right) - 1 \right]$$

$r^{\text{€}}$
EUR funding rate

which maybe higher or lower than the EUR RFR rate (ESTR). The basis is the difference between this implied funding rate and the RFR rate

$$\text{xCCy basis} = \underbrace{\frac{1}{\delta} \left[\left(\frac{S_0(1 + r^{\$}\delta)}{F(0, \delta)} \right) - 1 \right]}_{\text{(implied EUR funding rate)}} - r^{\text{€}}$$

$\delta = \frac{d}{360}$

$$\frac{1}{\delta} \left[\frac{S_0(1 + r^{\$}\delta)}{S_0 \left(\frac{1 + r^{\$}\delta}{1 + r^{\text{€}}\delta} \right)} - 1 \right] = \frac{1}{\delta} [r^{\text{€}}\delta] = r^{\text{€}}$$

Also note that if CIP holds, i.e., $F(0, \delta) = S_0 (1 + r^{\$}\delta) / (1 + r^{\text{€}}\delta)$, basis vanishes.

CSA Collateral Discounting

- To summarize the EUR discount factor for a maturity δ is given as

$$\frac{1}{(1 + r^{\text{€}} \delta)}$$

$$\frac{1}{(1 + (r^{\text{€}} + Z) \delta)}$$

EUR_EURCSA

EUR_USDCSA

$$z \rightarrow \delta$$

$$z' \rightarrow \delta'$$

where,

$$Z = \frac{1}{\delta} \left[\left(\frac{S_0(1 + r^{\$} \delta)}{F(0, \delta)} \right) - 1 \right] - r^{\text{€}}$$

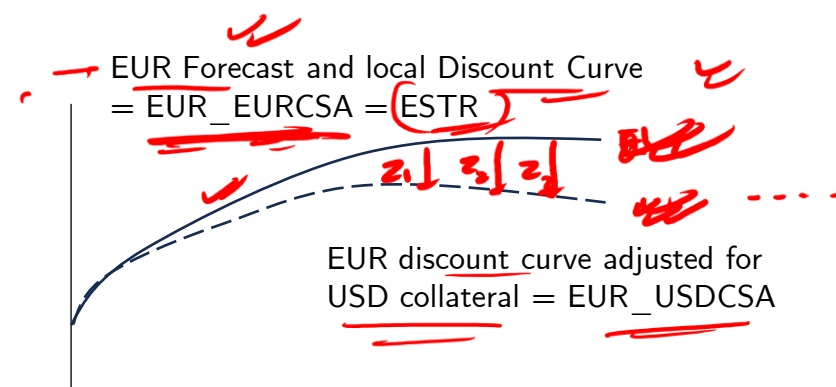
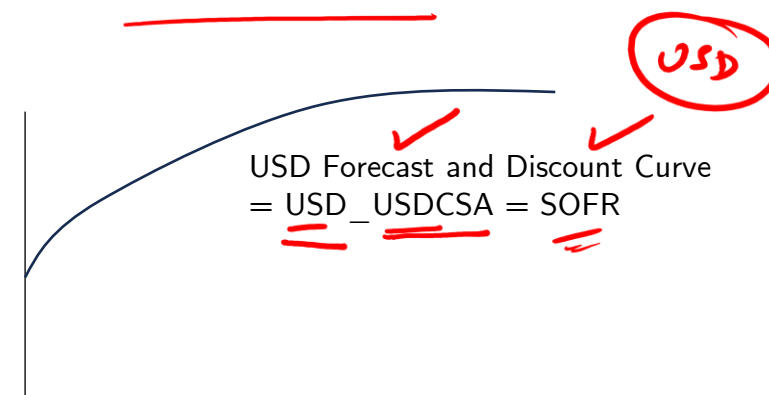
$$(r^{\text{€}} + z) = \frac{1}{\delta} [\cdot]$$

The presence of cross currency basis swap market alters the collateralized discount factor. With this modified discount factor one can generate market consistent forward curve.

Collateral Discounting (standard CSA)

Chart	Refresh	Export	Settings ▾	FX-Interest Rate Arbitrage							
Same Currency	Multi-Currency	Central Bank									
Currencies ▾	EUR	USD									
Spot Source	BGN										
Fwd Source	BGN										
ImPLY	EUR Yield										
FX Swap to EUR Depo											
		7) FX Swap		8) EUR Yield		9) USD Yield		EUR Implied Yield		Spread	
Term	Date	Bid	Ask	Bid	Ask	Bid	Ask	Bid	Ask	Bid	Ask
20) DEC	12/31/19	1.114404	1.114436	-0.4713	-0.4713	2.0919	2.0919	-0.6377	-0.6284	-0.1664	-0.1571
21) BJAN	01/02/20	1.115019	1.115080	-0.4724	-0.4724	2.0889	2.0889	-0.7776	-0.7600	-0.3052	-0.2876
22) 4M	01/13/20	1.116036	1.116076	-0.4777	-0.4777	2.0745	2.0745	-0.7979	-0.7874	-0.3202	-0.3097
23) 5M	02/12/20	1.118315	1.118365	-0.4800	-0.4800	2.0462	2.0462	-0.7477	-0.7372	-0.2677	-0.2572
24) 6M	03/12/20	1.120472	1.120537	-0.4944	-0.4944	2.0281	2.0281	-0.7070	-0.6956	-0.2126	-0.2012
25) MAR	03/31/20	1.121922	1.122001	-0.5030	-0.5030	2.0005	2.0005	-0.7114	-0.6988	-0.2084	-0.1958
26) BAPR	04/01/20	1.122000	1.122093	-0.5034	-0.5034	1.9987	1.9987	-0.7144	-0.6997	-0.2110	-0.1963
27) 9M	06/12/20	1.127287	1.127387	-0.5213	-0.5213	1.9061	1.9061	-0.7171	-0.7055	-0.1958	-0.1842
28) JUN	06/30/20	1.128605	1.128724	-0.5277	-0.5277	1.8846	1.8846	-0.7241	-0.7112	-0.1964	-0.1835
29) BJUL	07/01/20	1.128696	1.128814	-0.5279	-0.5279	1.8834	1.8834	-0.7263	-0.7135	-0.1984	-0.1856
30) 1Y	09/14/20	1.134065	1.134190	-0.5432	-0.5432	1.8104	1.8104	-0.7363	-0.7256	-0.1931	-0.1824
31) 2Y	09/13/21	1.158905	1.159524	-0.5929	-0.5929	1.6067	1.6116	-0.7924	-0.7611	-0.1995	-0.1682
32) 3Y	09/12/22	1.183450	1.184450	-0.5926	-0.5865	1.5125	1.5172	-0.7984	-0.7658	-0.2058	-0.1793
33) 4Y	09/12/23	1.208252	1.209748	-0.5702	-0.5641	1.4730	1.4779	-0.7889	-0.7534	-0.2187	-0.1893
34) 5Y	09/12/24	1.233200	1.234700	-0.5307	-0.5243	1.4619	1.4658	-0.7566	-0.7286	-0.2259	-0.2043
		FX Swap		EUR Yield		USD Yield		EUR Implied Yield		Spread	
Term	Date	Bid	Ask	Bid	Ask	Bid	Ask	Bid	Ask	Bid	Ask
	MM/DD/YY										
Legend											
* Represents Holiday ** Spot bid/ask replaced by mid.											
Australia 61 2 9777 8600 Brazil 5511 2395 9000 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000											
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2019 Bloomberg Finance L.P.											
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- We get the EUR implied rate term structure which serves as the input for EUR discounted curve adjusted for USD collateral



Collateral Discounting (non-standard CSA)

- What if the CSA is not in USD, but say in EUR? Then we need to adjust USD discount factor for EUR collateral i.e., USD_EURCSA. Non-standard CSA discounting curve can be generated using the simple idea that forward curve generated should not depend on which CSA is used. This is called as **FX Forward invariance**.

$$F(0, T)^{\text{EUR/USD}} = S(0)^{\text{EUR/USD}} \cdot \underbrace{\left(\frac{P(0, T)^{\text{EUR_USD_CSA}}}{P(0, T)^{\text{USD_USD_CSA}}} \right)}_{\text{(USD Collateral)}} = S(0)^{\text{EUR/USD}} \cdot \underbrace{\left(\frac{P(0, T)^{\text{EUR_EUR_CSA}}}{P(0, T)^{\text{USD_EUR_CSA}}} \right)}_{\text{(EUR Collateral)}}$$

Handwritten notes: "CIP" circled in red above the first fraction; "x CCy" circled in red above the second fraction.

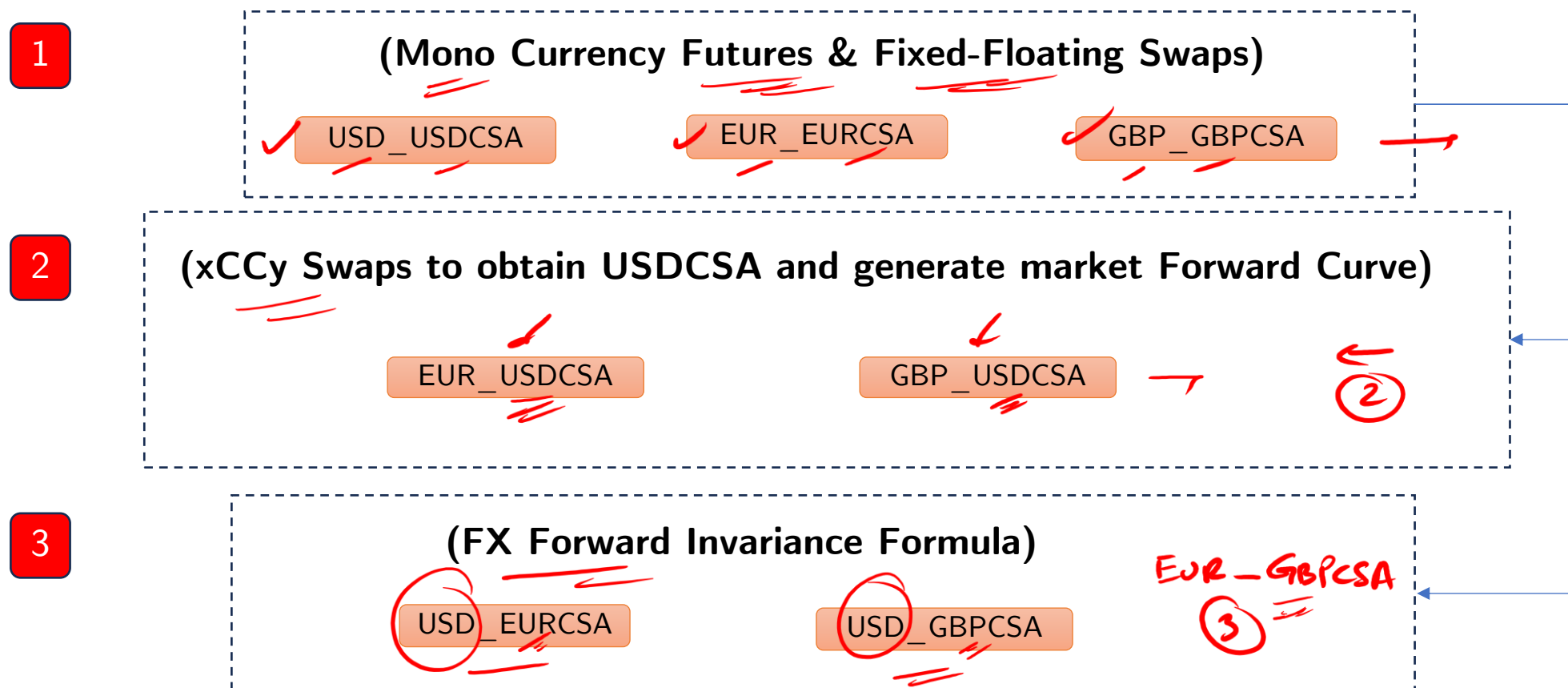
This equivalence leads directly to USD_EURCSA discount curve.

$$P(0, T)^{\text{USD_EUR_CSA}} = P(0, T)^{\text{EUR_EUR_CSA}} \cdot \left(\frac{P(0, T)^{\text{USD_USD_CSA}}}{P(0, T)^{\text{EUR_USD_CSA}}} \right)$$

Handwritten notes: "single EUR curve" pointing to the EUR_EUR_CSA term; "single USD curve" pointing to the USD_USD_CSA term; "USD_EUR_CSA" circled in red with a star and crossed out, indicating the result.

Multi-curve dependency chain

- Below is the hierarchy of curve construction from various instruments. If we are using bootstrapping, this order needs to be followed.



Optionality



- If the contract allow collaterals under multiple currencies, then one currency maybe cheaper to deliver compared to others. This is called **cheapest-to-deliver (CTD)**. Let's consider a simple example.

Delta has a derivative liability of \$100mm and \$100mm in cash. The terms of the CSA require ~~USD~~ cash (remunerated at RFR) or treasury bonds as collateral. The specified haircuts are 2% for maturity < 5 yr and 6% otherwise. The repo rate is RFR+8 bps. RFR is 0.75%. Delta can also borrow unsecured cash at rate 2.5%. Find the CTD option.

CTD

Delta's Strategy	Posted	Remuneration	Costs	1D Total
→ Post USD Cash	\$100,000,000.00 →	\$2,083.33	\$0.00	\$2,083.33
Borrow \$2mm and reverse-repo <5 yr bond	\$102,000,000.00	\$2,351.67	\$138.89	\$2,212.78
Borrow \$2mm and reverse-repo >5 yr bond	\$106,000,000.00	\$2,443.89	\$416.67	\$2,027.22

CTD discount Curves

- Because different collaterals earn different RFRs, it's possible to synthesize a CTD curve which generates the highest remuneration. The synthetic curve is called CTD discount curve. For example, consider two single-CSA discount curves exist one for JPY and one for USD. Then a multi-curve CSA discount curve can be created by taking the highest attainable rate each day. In the example below, one would post USD collateral for the first 2 days and JPY collateral for the next 2 days.

Date	JPY CSA	JPY CSA	USD CSA	USD CSA	JPY+USD	JPY+USD
	DF	O/N rate	DF	O/N rate	DF	O/N rate
t+0	1.000000	2.00%	1.000000	2.10%	1.000000	2.10%
t+1	0.999944	2.05%	0.999942	2.10%	0.999942	2.10%
t+2	0.999888	2.15%	0.999883	2.10%	0.999883	2.15%
t+3	0.999828	2.20%	0.999825	2.10%	0.999824	2.20%
t+4	0.999767	etc...	0.999767	etc...	0.999763	etc...

CTD DF